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An artificial intelligence-enabled industry classification and its interpretation

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Abstract

Purpose – To overcome the shortcomings of traditional industry classification systems such as the Standard Industrial Classification Standard Industrial Classification, North American Industry Classification System North American Industry Classification System, and Global Industry Classification Standard Global Industry Classification Standard, the authors explore industry classifications using machine learning methods as an application of interpretable artificial intelligence (AI).

Design/methodology/approach – The authors propose a text-based industry classification combined with a machine learning technique by extracting distinguishable features from business descriptions in financial reports. The proposed method can reduce the dimensions of word vectors to avoid the curse of dimensionality when measuring the similarities of firms.

Findings – Using the proposed method, the sample firms form clusters of distinctive industries, thus overcoming the limitations of existing classifications. The method also clarifies industry boundaries based on lower-dimensional information. The graphical closeness between industries can reflect the industry-level relationship as well as the closeness between individual firms.

Originality/value – The authors' work contributes to the industry classification literature by empirically investigating the effectiveness of machine learning methods. The text mining method resolves issues concerning the timeliness of traditional industry classifications by capturing new information in annual reports. In addition, the authors' approach can solve the computing concerns of high dimensionality.

Keywords Industry classification, Text mining, Machine learning, Autoencoder, Dimensionality reduction, Firm similarity

Paper type Research paper

1. Introduction

Industry classification is a useful means of categorizing firms so that similar businesses are grouped to identify trends within an industry. Most business databases provide industry classifications and allow searches using keywords in a business description. The government also needs accurate industry classification for balanced economic development or for fostering certain industries. Thus, industry classification is an important issue for practitioners and scholars in business and economics (Davis and Duhaime, 1992; Hoberg and Phillips, 2016).

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However, traditional industry classification systems such as the Standard Industrial Classification (SIC), North American Industry Classification System (NAICS), and Global Industry Classification Standard (GICS) have certain limitations. As the contemporary economy changes rapidly, a company may enter or exit an industry much faster than ever. Moreover, a new industry could emerge and grow at a faster pace. This issue of timeliness might be critical when managers use industry classifications to understand their business areas or decide on strategic directions.

As advances in computing power enhance the ability of scholars to deal with complex calculations, researchers in information systems have adopted a variety of approaches to classify industries for grouping homogeneous firms (Fang *et al.*, 2013; Chowdhuri *et al.*, 2014; Yang *et al.*, 2019; Xu *et al.*, 2020). Among many academic works related to alternative industry classifications, Hoberg and Phillips (2016) develop the text-based network industry classification (TNIC) derived from the firms' business descriptions of 10-K reports. Using the cosine similarity measure to identify how similar a firm's business description is compared with all other firms, they construct 300 industries based on the results of a pairwise comparison to categorize a peer firm as a rival and/or a competitor. However, the pairwise comparison approach is limited because it provides only one-dimensional information that might only represent a firm-to-firm relation. Moreover, the pairwise comparison approach cannot infer the overall industry closeness and relationship, although they validate the across-industry variation of their final clusters.

To present an effective method of industry classification, it is necessary to take into account the issue of high dimensionality arising from text analyses. As pointed out by Radovanović *et al.* (2010), the cosine similarity measures from high-dimensional word vectors may not be successful in discriminating features. That is, as dimensionality increases, the mean becomes constant and variance converges to zero resulting in the concentrated cosine similarity measures. Because our text data extracted from financial statements consist of huge word vectors, we need to address such high-dimensionality problems. We address this issue using a machine learning technique. Artificial intelligence (AI) or machine learning is transforming the way people approach or understand financial risk management (Aziz and Dowling, 2019). AI is defined as intelligence demonstrated by machines to the level of intelligence equivalent to humans (Shieber, 2004). Machine learning using text mining is a core part of AI, as it involves learning from data. Therefore, this paper presents a new method for industry classification using AI with textual analysis.

We use a method that employs a deep autoencoder to reduce the dimensionality of word vectors from the business description of annual 10-K reports. In comparison to previous research, this approach mitigates the high dimensionality problem of the clustering method based on the cosine similarity measure. Our results show that dimensionally reduced features can sufficiently capture the relationship between firms, as well as between industries. We apply a spherical k-means algorithm to cluster industries using reduced features from the coded layer output of the autoencoder.

Our proposed industry classification method can overcome the shortcomings of traditional industry classifications such as SIC, NAICS and GICS. Hoberg and Phillips (2016) point out that existing fixed industry classifications such as SIC or NAICS may have time-fixed location restrictions. That is, existing fixed industry classifications might not reflect timely information of firms' industry groups. They develop text-based network industry classification to allow product market definitions to change every year by reflecting information in10-K product descriptions. Similar to the text-based classifications suggested by Hoberg and Phillips (2016), our industry classification system dynamically reflects new information in annual 10-K reports as it can compute new industry classifications based on companies' business descriptions every year. By contrast, the SIC code aggregates firms based on information regarding the sale of end products and similar production processes (Chan *et al.*, 2007). Despite the wide application of

SIC codes in previous empirical studies, many researchers have questioned their usefulness as a standard for industry classification (Bhojraj *et al.*, 2003; Kile and Phillips, 2009). For example, Walker and Murphy (2001) point out that the SIC classification may not sufficiently reflect shifts in main products, business processes, and emerging markets because the system mainly emphasizes manufacturing operations relative to service processes. Second, our industry classification uses an autoencoder and provides a visual representation of the industry classification, which contains information about firm-to-industry and industry-to-industry relations as well as firm-to-firm relations. Thus, our method can overcome the limitations of previous text-based industry classifications.

Moreover, our industry classification does not have the high dimensionality problem that might exist when measuring the distance between firms because autoencoders can reduce the dimension of word vectors. By contrast, Hoberg and Phillips (2016) compute the distance between firms with word vectors larger than 60,000 dimensions to classify the industry. However, Aggarwal *et al.* (2001) show that the distance measure (the concept of proximity) may not be qualitatively meaningful in a high-dimensional space. The cosine similarity measure is mathematically identical to the L2-normalized Euclidean distance. This implies that similarity measures by high-dimensional vectors cannot escape the curse of dimensionality because the space of the cosine similarity measure remains highly dimensional (Skillicorn, 2012).

The remainder of this article is organized as follows. First, we review related literature. Then, we describe our data and methodology and present the results. Finally, we summarize our study and discuss its contributions to the literature.

2. Literature review

2.1 Industry classifications

The US government first developed the SIC in 1937. In 1997, NAICS was developed to represent the industries of the United States, Canada, and Mexico. Subsequently, the Standard and Poor's and Morgan Stanley Capital International established GICS, which is designed for financial analysts and investment managers to determine which firms are financially comparable. In academia, Fama and French (1997) establish their own classification schemes by reclassifying existing SIC codes to construct industry portfolios. Their scheme is called Fama–French (FF) Industry Classifications, which has fixed industries ranging from 5 to 49. This classification system is prevalent in financial and accounting research. Bhojraj *et al.* (2003) and Chan *et al.* (2007) compare the reliability and accuracy of these systems and find that GICS in many cases outperforms SIC and NAICS.

Recent research in the field of information systems has provided an alternative way to classify industries, as researchers are now able to handle large amounts of computing. Fang *et al.* (2013) criticize the SIC and NAICS, as these assume a binary relationship and might not measure the degree of similarity. To overcome these issues, the authors use latent Dirichlet allocation (LDA) to propose an industry classification methodology that is based on business descriptions. Chowdhuri *et al.* (2014) propose an ontology-based framework that provides interoperability between different eXtensible Business Reporting Language (XBRL) filings for XBRL mapping and decision-making. Yang *et al.* (2019) use a graph similarity metric, combined with a spectral clustering algorithm, to quantify the similarity of financial disclosures. Xu *et al.* (2020) present an industry classification method by implementing labor mobility network using data sets of LinkedIn profiles.

2.2 Text mining methods and high dimensionality issues

Text mining methods have been explored both in research and practice. Classical text mining methods typically employ techniques that transform texts into vectors (Aggarwal and Zhai, 2012). Other types of classification techniques such as random forests, decision tree classifiers, rule-based classifiers and conditional random fields have been used (Xu *et al.*, 2012; Harrag *et al.*, 2009; Lafferty *et al.*, 2001).

An autoencoder is a dimensionality reduction technique based on the deep neural network first introduced by Baldi and Hornik (1989). Hinton and Salakhutdinov (2006) improve the model to make it generative by applying the greedy layer-wise pre-training technique. Hinton and Salakhutdinov (2006, p. 504) describe an autoencoder as "a nonlinear generalization of PCA that uses an adaptive, multilayer encoder network to transform the high-dimensional data into a low-dimensional code and a similar decoder network to recover the data from the code."

An autoencoder can transform high-dimensional data into low-dimensional data. Hence, a well-trained autoencoder represents the original data closely while ignoring its signal noise. An autoencoder contains a small central layer (called "code") that connects two bigger multilayer neural networks on both sides, similar to a butterfly's wings such that one wing is called an encoder and the other is called a decoder. Because the input and output layers have the same number of nodes, an autoencoder receives high-dimensional input data at one end and reproduces them at the other end. Because an autoencoder does not predict a target value but only reconstructs the input data, it performs unsupervised learning and does not require labeled data.

The first step to train an autoencoder is to assign random weights to the two networks at each side of a smaller central layer (code). The next step is to minimize the difference between the original data plugged into the input layer and the reconstructed version produced from the output layer. In practice, Hinton and Salakhutdinov (2006) recommend performing procedures that pre-train two adjacent layers (restricted Boltzmann machine) before proceeding to layer-by-layer estimation because multiple hidden layers are largely untrainable. Once the pre-training stage is completed, a full autoencoder network is unrolled and then fine-tuned. Estimating an autoencoder network is simply expressed as

$$f : X \rightarrow Y, g : Y \rightarrow X, \ \widehat{X} = (g \circ f)X, \ [\widehat{f}, \ \widehat{g}] = \underset{(f,g)}{\operatorname{argmax}} d(X, \ \widehat{X}).$$

where X is an input space; f is an encoder; g is a decoder; and $d\ (+)$ indicates the distance metric. In summary, an autoencoder is unsupervised machine learning, where the number and shape of the output node are the same as the number and shape of the input node.

An autoencoder is fundamentally a dimension-reduction technique that facilitates noise reduction, classification, visualization, data storage, and a generative approach. Dimension reduction aims to represent data from a high-dimensional space to a low-dimensional space. Principal component analysis (PCA) is the best-known dimension-reduction technique. However, an autoencoder outperforms PCA in classifying and interpreting clusters in the original data (Hinton and Salakhutdinov, 2006). Furthermore, an autoencoder outperforms latent semantic analysis, a well-known document retrieval method based on PCA, in computing the cosine similarity between two vectors (Deerwester *et al.*, 1990). An autoencoder also outperforms local linear embedding, a recent nonlinear dimensionality reduction algorithm (Roweis and Saul, 2000).

While Hoberg and Phillips (2016) compute the cosine similarities to use the word vectors that contain whole unique words, we calculate the similarities based on the reduced word vectors extracted from the autoencoder. In other words, we use the reduced features to divide firms into several groups or industries by using a spherical *k*-means clustering algorithm suitable for the vector space model of the corpus. Prior literature shows that our industry classification of firms is quite interpretable and intuitive despite the small loss of information because the autoencoder uses the sparsity of the hidden layers when the model is trained and applied to a nonlinear problem beyond the PCA approach.

In addition, an autoencoder can detect anomalies well. For example, some data points could show large reconstruction errors, which are regarded as anomalies while an autoencoder reconstructs the original data. To extend this intuition, Misra *et al.* (2020) use an autoencoder to detect anomalous credit card transactions and deem them fraudulent. Their approach is two-step: an autoencoder (1) reduces the dimensions of transaction attributes, and (2) classifies transactions into clusters based on the feature vector of reduced dimensions. Note that this approach is essentially the same as our industry classification method.

Furthermore, the anomaly detection technique can be used to identify mispriced financial assets (e.g. undervalued or overvalued stocks; Gu *et al.*, 2021). Hence, intuitively, by combining mispriced assets with a small number of communal stocks that tend to co-move with their originals, investors can generate excess portfolio performance over a benchmark. Indeed, Heaton *et al.* (2017) define a target return that follows a benchmark index, but with downside protection and then construct "deep portfolios" to replicate the target return, that is, an enhanced benchmark index. The deep portfolio comprises anomalous and communal stocks. Meanwhile, Nakano and Takahashi (2020) construct factors (the central code in an autoencoder) that present downside protection and then replicate the factors with dynamic hedging strategies to enhance portfolio performance.

3. Data and methods

3.1 Data and sample construction

We use web-crawling algorithms to collect 10-K annual reports filed with the US Securities and Exchange Commission (SEC) from 2013 to 2016. We then extract the business description section of each document to arrive at 21,631 10-K reports. The 10-K business description section is legally required of all firms by item 101 of Regulation S-K, which instructs firms to describes the core products they offer to a market.

We use the Compustat historical segment database to obtain the corresponding SIC codes for each firm. The segment information provided in the Compustat segment data is self-reported by the reporting firms. Since 1976, US firms have been required to provide segment information for each industry segment that represents 10% of firm revenues [1]. The reporting period and financial activity period of the Compustat segment data may contain redundancies because each firm is required to report the information for the past three years at every reporting period. Thus, we only keep the latest year for each report. For example, in 2015, a firm reported its segment information for 2013, 2014 and 2015. In this case, we keep only the 2015 information when the reporting year is 2015. We use the primary SIC code for the segment information of each firm and year, removing observations when a firm is missing a primary code. We merge the Compustat data with the 10-K annual report document that we have crawled by using the SEC Central Index Key. After merging the SIC code by individual firm and year, our sample consists of 14,560 firm-year observations.

3.2 Bag-of-words representation

We process the 10-K business descriptions to construct bag-of-words and word vectors. The words from 10-K business descriptions can contain information about certain business processes and products that each firm offers in the markets. The underlying rationale is that firms classified in the same industry are expected to use more similar words to describe their businesses and products. Table 1 illustrates the 20 most used words extracted from the business descriptions of sample firms that belong to different industries. For example,

Firm 1: SANDISK CORP (SIC code: 3572) Business: Flash Memory Storage Core words: memory(67), product(52), technology(44), storage(36), market(31), device(31), solution(28), NAND(26), flash(24), drive(20), manufacturer(19), design(19), corporation(18), venture(18), card(18), president(17), data(16), wafer(16), cost(15), year(15)
Firm 2: SCHEIN (HENRY) INC (SIC code: 5047) Business: Healthcare Distribution Core words: health(102), product(89), care(65), service(62), state(47), customer(47), law(44), practice(41), president(40), business(40), distribution(37), sale(35), drug(33), act(30), vice(30), practitioner(26), officer(25), technology(24), order(23), management(23)
Firm 3: IMPAC MORTGAGE HOLDINGS INC (SIC code: 6162) Business: Long-term Portfolio Core words: mortgage(174), loan(159), origination(53), portfolio(49), service(45), estate(34), operation(33), channel(33), Mae(30), interest(30), correspondent(29), lending(27), rate(27), credit(21), security(21), broker(20), sale(20), borrower(19), seller(18), act(17)
Firm 4: TENGASCO INC (SIC code: 1311) Business: Oil and Gas Core words: company(200), gas(96), methane(44), production(43), well(37), oil(36), agreement(33), Hoactzin(30), pipeline(28), management(27), program(26), interest(25), sale(25), property(24), operation(23), swan(23), report(22), project(21), price(21), field(21)
Firm 5: ACCELRYS INC (SIC code: 7372) Business: Software Development Core words: product(82), software(54), customer(46), platform(34), development(32), service(29), data(27), solution(26), process(24), market(24), acquisition(24), research(24), enterprise(23), industry(23), organization(22), quality(19), informatics(19), system(18), management(18), information(17)

Note(s): This table illustrates the top 20 words frequently used in the business descriptions of sample firms. The value in parentheses represents the number of the frequency of each word. For example, SANDISK CORP uses the word "memory" 67 times in the business description part of the 10-K annual report in 2013

Table 1. Frequent words in the business descriptions

SANDISK used the word "memory" 67 times in the business description section of their 2013 10-K annual report to describe their business and products.

For a generalized form, we use a bag-of-words representation to convert the business description into a vector form. The bag-of-words representation is widely used in information retrieval and text mining (Singhal, 2001). The bag-of-words is a vector where each component is the frequency of a given word found in a set of documents. The representation assumes that words appear independently, and the order of the words is immaterial. The number of words, thus, corresponds to the dimensions in vector space, and each document then becomes an individual vector containing non-negative values on each dimension.

In this study, a conventional text mining process is applied to structure a bag-of-words or a set of unique words from preprocessed documents. To remove commonly used words, we focus only on nouns and proper nouns that appear in no more than 20% of the business descriptions. A threshold of 20% is selected following similar criteria in Hoberg and Phillips (2016). They indicate that minor modifications of the threshold do not significantly affect the main firm results. We remove geographical words, such as country names and names of the world's popular cities. Business description texts containing fewer than 20 unique words are excluded because they lack unique information about firms. We assume that unique words exist in the documents reported by the firms in the training samples.

After excluding common words, we construct a word vector W containing the 2,000 most frequently used unique words in all the documents in our sample. We take the vector W and convert the individual business descriptions of firms to the corresponding word vectors. The words in the business description of each firm i can be represented by the 2,000-dimensional binary (coded) vector V_i. Each element v_{ij} of vector V_i equals one if the given bag-of-words

W contains the word element from the word list of the individual document, and zero otherwise. Figure 1 depicts a schematic illustration of the codification of word vectors. For example, when the word "memory" appears in the business description of firm i and exists in the first row (w_1) of the bag-of-words vector W, the element v_{i1} is coded as a value of one, as the bag-of-words has the word "memory" at location w_1.

3.3 Dimensionality reduction

Hoberg and Phillips (2016) use a cosine similarity measure to compute the pairwise similarity between two normalized vectors containing all unique words. They focus on nouns and proper nouns with less than 25% of all product descriptions by excluding geographical words such as cities, countries, and states. They report that typical firms use approximately 200 unique words. They then construct each firm's word vector by assigning the value of one for the word contained in the vector of whole unique words, and zero otherwise. In a similar way, we extract unique words from the product descriptions of 10-K reports. Figure 2 shows

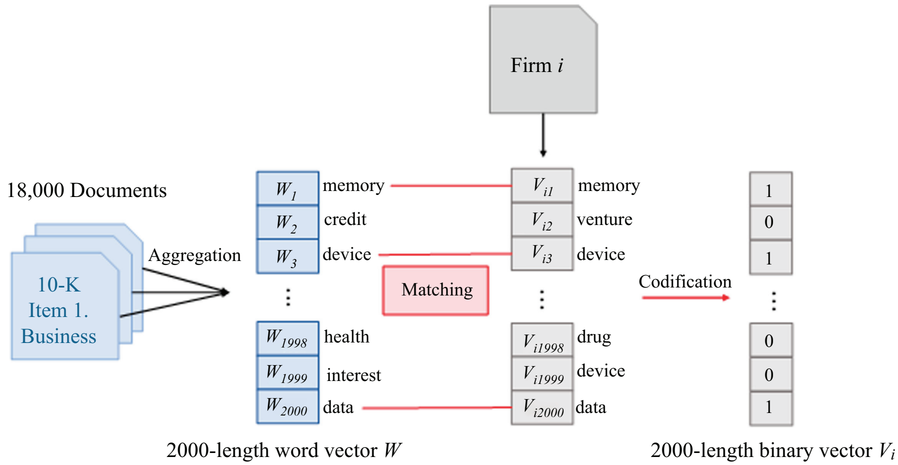


Figure 1.
Word vector codification using the bag-of-words

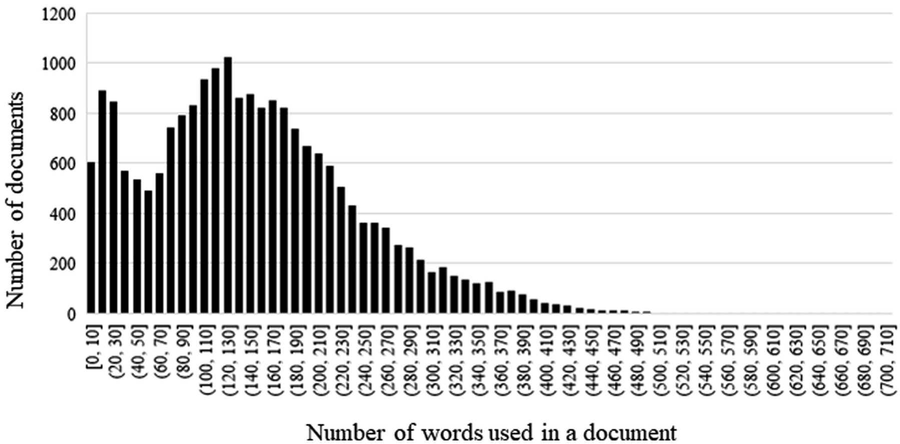


Figure 2.
Distribution of the number of words used in 10-K business descriptions

the frequencies of the number of unique words in each business report in our sample. Our sample shows that 100 to 200 unique words are frequently used in business descriptions, with an average of 151 unique words being used.

The cosine similarity formula can be directly represented in terms of the normalized Euclidean distance. However, the cosine similarity measure based on the Euclidean distance cannot mitigate the problem of clustering when the vector space has high dimensions. For example, Radovanović *et al.* (2010) point out that the expectation of pairwise cosine similarity measure might become constant and the variance of the measure might shrink to zero as dimensionality increases. Thus, the high-dimensional vector space may lead to the curse of dimensionality when computing the cosine similarity measure.

To mitigate the high dimensionality problem of the cosine similarity measure, we employ a dimensionality reduction method using an autoencoder. The autoencoder is an unsupervised machine learning technique in which the number and shape of the output node are designed to be the same as those of the input node. An autoencoder is first applied by implementing a PCA-like dimensionality reduction using the backpropagation method in a shallow neural network. The model consists of two parts: encoding and decoding layers. The model can reduce the number of hidden (encoded) nodes while minimizing the difference between the input vector and the reconstructed output vector at the end of the model.

Figure 3 illustrates the autoencoder scheme used in this study. We implement the autoencoder regime as a dimensionality reduction technique to create a low-dimensional vector containing industry information where the firms are classified. The dimension of the 2,000-word vector can be reduced to 10 dimensions, which are used as the features for the clustering step. The layer structure of our learning model is 2,000–500–125–10–125–500–2,000 with the rectified linear unit (ReLU) as an activation function of each layer, except for the last of the encoding and decoding layers. The last encoding layer uses a linear function that unbinds the values of the reduced vector. The sequence of numbers, 2,000–500–125–10–

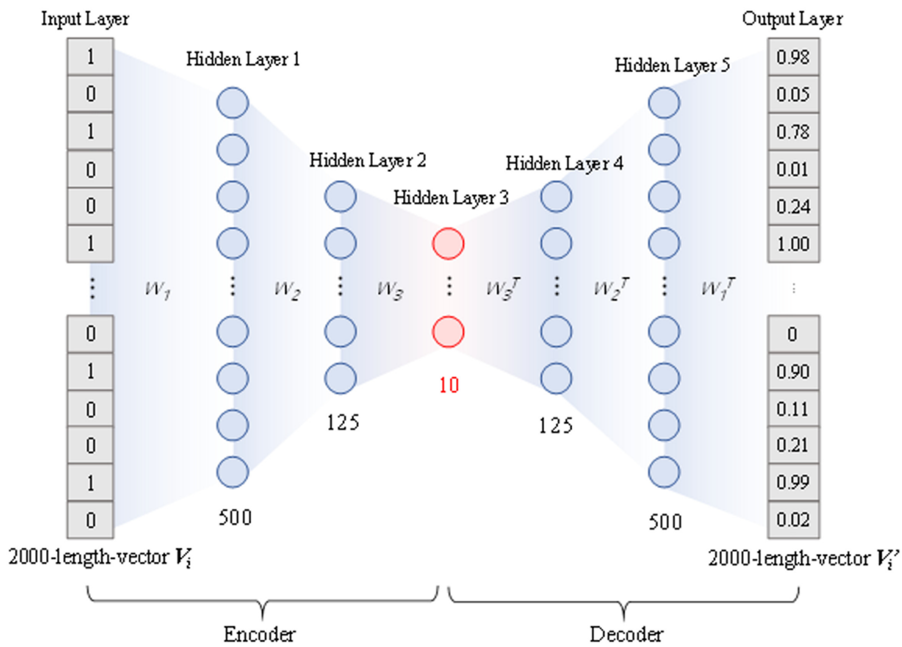


Figure 3.
Schematic structure of autoencoder

125–500–2,000, refers to the number of nodes in each layer of the autoencoder system. The numbers prior to the code (2,000–500–125) denote the encoding part in which the autoencoder reduces the dimensions from 2,000 to 500, from 500 to 125, and finally from 125 to 10 dimensions. The autoencoder system then reconstructs each layer from 10 to 2,000, similar to the encoding part. We minimize the value of the binary cross-entropy as a loss function used in the training stage. To measure the value of the loss function, the sigmoid function is selected when computing the loss function at the end of the decoding (output) layer. This structure sufficiently discriminates and explains information about the 2,000-word-long vectors, although we modify the number of the hidden nodes.

3.4 Spherical k*-means clustering*

Standard *k*-means clustering minimizes the mean-squared error of the Euclidean distance within the clusters. A well-known underlying probability distribution for the standard *k*-means algorithm is the Gaussian distribution. In this study, we instead use a spherical *k*-means clustering algorithm, which is a suitable clustering algorithm for the vector space model of the corpus because the direction of the word vector is more important than the magnitude. For high-dimensional data such as text documents, cosine similarity is superior to Euclidean distance (Strehl *et al.*, 2000). The spherical *k*-means algorithm maximizes the average cosine similarity measure within the clusters. The main difference from standard *k*-means is that the re-estimated mean vectors must be normalized to unit length, and the underlying probabilistic models are not necessarily the Gaussian distribution. We cluster the reduced features into several industries by applying the spherical *k*-means algorithm. The results of the spherical *k*-means clusters are directly compared to the SIC, NAICS and GICS codes, and Hoberg and Phillips's (2016) 300 fixed TNIC codes. We also compute the cosine similarity measure between the firms based on the reduced features. We validate the proposed method by comparing within and across the variations in terms of profitability and risk levels.

4. Results

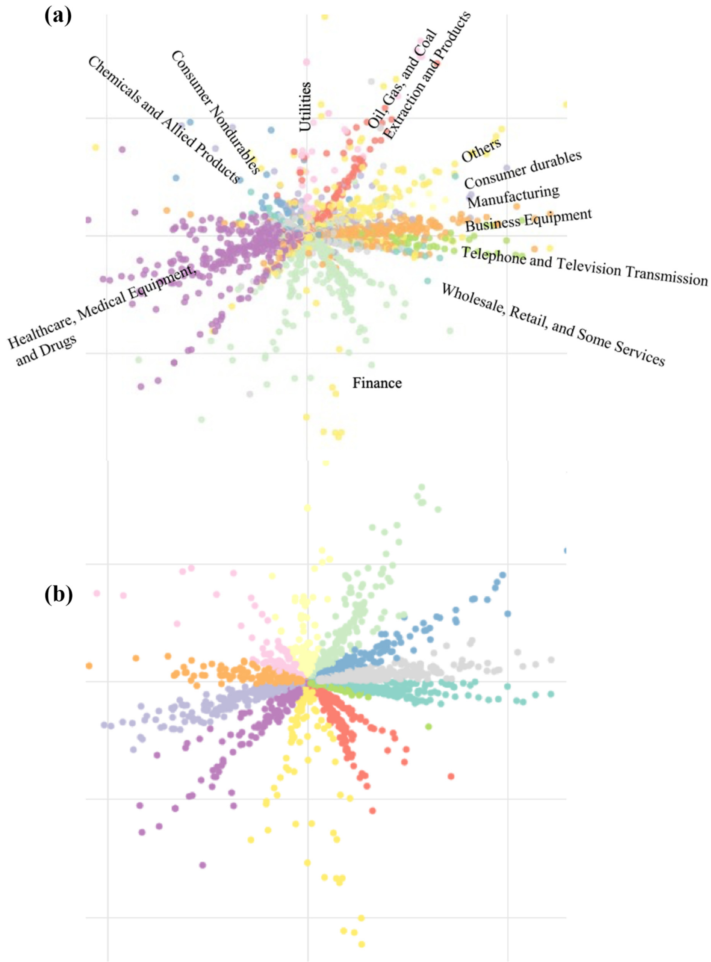
4.1 Graphical analysis

For a two-dimensional visual representation, we modify the number of hidden nodes in the last encoding layer. The modified layer structure is 2,000-500-125-2-125-500-2,000. The structure differs from the previous model only in that the number of nodes at the coded layer is two.

Figure 4 shows the two-dimensional scatter plots of the latent values from the last encoding layer after training the autoencoder. For a simple comparison, we classify our sample into 12 categories, which are analogous to the Fama-French 12 industry classifications (FF12). Fama and French (1997) reclassify conventional SIC codes into 48 industry groupings and add more groupings such as 5, 10, 12, and so on later. Although they use existing SIC codes to construct their classifications, the FF industry classifications are designed to share common risk characteristics. For example, they assign group 4 as the industry "Energy, Oil, Gas, and Coal Extraction and Products," which includes firms in SIC 1200–1399 and 2900–2999 although the first two digits of the SIC codes are different.

Figure 4(a) illustrates that firms labeled as FF12 are clustered on a single spike from the origin, which indicates that firms in the same industry have similar directions with different vector norms. For example, the range of the SIC codes colored yellow–green in Figure 4(a) is 6000–6999, which are financial firms. This result is consistent with the Hoberg and Phillips'(2016) clustering results because they use the cosine similarity to measure the degree (distance) between two business descriptions.

Figure 4(b) shows the results of the spherical *k*-means clustering. The clustering results indicate that financial firms and medical-related firms are divided into several sub-industries.



Note(s): The coordinate of each dot represents the text-based classification after extracting two dimensional codes from the autoencoder. The color of each dot in Figure 4(a) represents the classification based on Fama-French 12 industry classifications. Figure 4(b) indicates the spherical clustering result using the reduced features of the coded layer

Figure 4.
The scatter plot of text-based classifications

Some firms happen to be far from their industry groups (spike) according to the FF12 classification. This result indicates that these firms might use different words to describe their business processes and products when compared with other firms that belong to the same industry based on the SIC code or the FF12 classification. These firms may have improperly reported their business description or their SIC codes are wrongly assigned because they may have recently changed their major products or business processes through diversification and pivoting.

The proposed method can resolve the latter problem by identifying more appropriate industries based on the list of words appearing in the firm's business descriptions. Hoberg and Phillips (2016) exemplify that their text-based measures can reveal that firms in the newspaper publishing and printing industry (three-digit SIC is 271) and firms in the radio and broadcasting stations industry (three-digit SIC is 483) are quite similar groups because both industries can serve the same customers who want advertising. However, the fixed SIC codes may treat these firms differently because of the different starting digits in the SIC codes. Thus, text-based measures based on business descriptions can correctly specify the industry to which each firm belongs.

4.2 Case studies

This section illustrates that our text-based classification can capture the firms' industry by reflecting their current business areas. That is, our text-based classifications can present timely industry classification that reflects the firms' products or services described in their annual reports. For example, Netflix was originally involved in the video rental business but has recently expanded its business area to provide online TV programs and movies and to create new content. Netflix is still assigned a SIC code of 7841 (video tape rental), which is quite out of date. However, our text-based classification shows that Netflix is categorized in a similar group as Dish Network, DreamWorks, TiVo, and Outdoor channels. Thus, our classification seems to reflect the current business areas of companies better than the SIC codes.

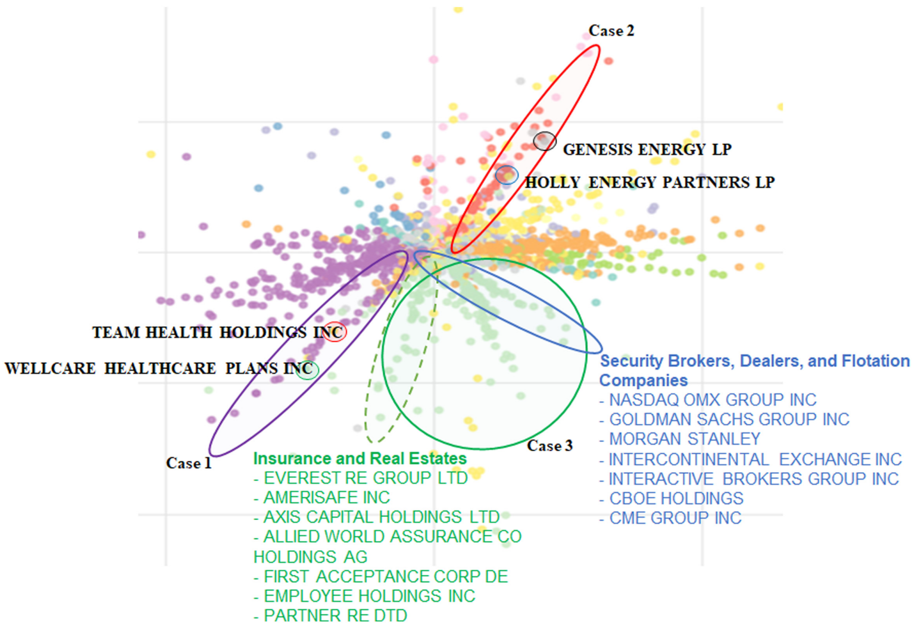
We also provide two cases to illustrate that traditional methods may mislead industry classification. Table 2 shows that our proposed method classifies similar firms into the same category, but the SIC codes and the FF12 classification may have different results. Table 3 lists common words belonging to each case to show that our proposed method is suitable for identifying each firm's industry. Figure 5 illustrates that firms with different SIC codes might be grouped into the same category based on our proposed method. Because our classifications are based on each firm's actual product descriptions, we can detect potential peer firms that offer related products, although they are not currently connected.

Table 2. Comparison of clustering results with SIC codes and Fama–French 12 classification

Firm name	SIC code	FF12 Classification	Clustered code
Case 1 – Healthcare, Medical Equipment, and Drugs			
TEAM HEALTH HOLDINGS INC	7363	12 Others	11
WELLCARE HEALTH PLANS INC	6324	11 Money	11
SELECT MEDICAL HOLDINGS CORP	8069	10 Health	11
SYMBION INC TN	8011	10 Health	11
LHC GROUP INC	8082	10 Health	11
LIFEPOINT HOSPITALS INC	8062	10 Health	11
TENET HEALTHCARE CORP	8062	10 Health	11
AMN HEALTHCARE SERVICES INC	8090	10 Health	11
HCA HOLDINGS INC	8062	10 Health	11
Case 2 – Oil, Gas, and Coal Extraction and Products			
GENESIS ENERGY LP	5171	9 Shops	4
CROSSTEX ENERGY LP	5172	9 Shops	4
HOLLY ENERGY PARTNERS LP	4613	12 Others	4
GULFPORT ENERGY CORP	1311	4 Energy	4
CONTINENTAL RESOURCES INC	1311	4 Energy	4
UNIT CORP	1311	4 Energy	4
MID CON ENERGY PARTNERS LP	1311	4 Energy	4
CHEVRON CORP	2911	4 Energy	4

Unique words out of 2000 words in the bag-of-words	Frequency
Case 1 – Healthcare, Medical Equipment, and Drugs	
Hospital, billing, physician, Medicaid, productivity, patient, submission, reimbursement, Medicare, referral	9 out of 9 documents
Beneficiary, methodology, recruitment, length, abuse, accountability, authorization, accreditation, CM, associate, prohibition, utilization, therapy, transition, employer, sanction, eligibility, safeguard, notification, fraud, worker, HIPPA (Health Insurance Portability and Accountability Act), spending, portability, admission, antikickback, update	8 out of 9 documents
Case 2 – Oil, Gas, and Coal Extraction and Products	
Crude, commodity, pipeline, hydrocarbon, petroleum, transport	8 out of 8 documents
Carrier, cleanup, liquid, pollution, barrel, index, discharge, mile, tank, basin, emergency, exploration, drilling, commerce, injection, FERC(Federal Energy Regulatory Commission), shale, formation, greenhouse, dioxide, emission, gathering, fuel	7 out of 8 documents

Table 3. Most frequent words within the same industry



Note(s): The coordinate of each dot represents the text-based classification after extracting two dimensional codes from the autoencoder. The color of each dot represents the classification based on SIC code

Figure 5. Graphical comparison between the text-based classifications and SIC codes

Table 2 and Figure 5 illustrate two cases. The first case (Case 1) is related to the healthcare, medical equipment, and drugs industry. The second case (Case 2) is related to oil, gas, and coal extraction and products. The colors of the dots represent the FF12 classification, whereas our classification can be identified by the direction of each spike.

For Case 1 of Table 2, the SIC code of the firm “TEAM HEALTH HOLDINGS INC” is assigned as 7363, and the FF12 classification is 12. Another firm “WELLCARE HEALTH

PLANS INC" has the 6324 SIC code, and the FF12 classification code is 11. The former firm provides outsourced healthcare professional staff and administrative services to hospitals, while the latter provides managed care services and government-sponsored healthcare programs. These two firms may provide similar services in terms of healthcare products. This situation is correctly reflected in our classification system of 11, assigning the two firms to the same industry. However, the former firm is classified as 11 (Money) and the latter, 12 (Others) by the FF 12 classification. The other firms clustered with these two firms in Case 1 have SIC codes starting at 80 and an FF classification of 10. In Figure 5, firms related to healthcare products and services are included in the purple ellipse. Most firms (each point) in the purple ellipse have the same color, but some firms have different colors, which indicate other categories based on the FF12 classification.

Case 2 is related to the energy and mining industry, including oil, gas, and materials. The first two firms of Case 2 in Table 2, GENESIS ENERGY LP and CROSSTEX ENERGY LP, belong to the "Shops" industry according to the FF12 classification and "Petroleum stations and terminals" based on the SIC codes. However, our proposed method assigns these two firms to the same energy-related industry coded as 4. Moreover, Figure 5 illustrates that the two firms are closer to "Energy" firms in terms of the products and business processes, and many firms involved in the oil, gas, and coal extraction and products are identified by the red ellipse.

Table 3 shows the list of words that overlap in the documents reported by the firms in Cases 1 and 2. For example, Table 3 shows that for Case 1 nine firms related to healthcare, medical equipment, and drugs use the words "hospital," "billing" and "Medicare," which appear nine times in the 10-K annual reports. Moreover, the words "abuse," "therapy" and "HIPPA" appear eight times across the nine firms. For Case 2, the words "crude," "commodity" and "pipeline" appear eight times and the words "carrier," "cleanup" and "liquid" appear seven times in the documents of eight firms.

Figure 5 also highlights another issue related to the sub-industry (Case 3). Three or four groups with spikes are observed within the financial industry, as depicted by the green dashed circle. In particular, our method divides the financial industry into three or four sub-industries, while the FF12 classification or SIC codes classify these firms into a single category. Each spike represents different clusters based on words that appeared in their business descriptions, while these firms are all related to money or the financial industry. Thus, our method can reveal the closeness of these firms through visualization by dividing similar firms into subgroups (sub-industries). Among the subgroups, most of the firms in the dashed-green circle (ellipse) are related to insurance such as EMPLOYEE HOLDINGS and ALLIED WORLD ASSURANCE CO HOLDINGS. The financial firms in the blue circle are related to the security broker and dealer business such as GOLDMAN SACHS, MORGAN STANLEY, CBOE HOLDINGS, and CME GROUP. While both insurance and brokerage businesses belong to the financial industry, the characteristics of these businesses might be different. The business scope of insurance firms may be related to the healthcare business, which is represented by the purple circle (Case 1). Moreover, many trading firms employ advanced technology in their businesses. For example, high-frequency trading firms or major stock exchanges use advanced networks or intensive technology to maintain their core businesses. Thus, the security brokers and dealers in the blue circle are possibly more related to the IT industry based on their business descriptions.

4.3 Comparison based on within or cross standard deviations

Companies in the same industry may have similar business structures and be exposed to similar risks. An economic shock can affect all firms in a certain industry, either negatively or positively. For example, the COVID-19 pandemic can affect the aviation or travel industries negatively and the electronic education industry positively. Thus, if our method selects similar firms belonging to the same industry, the variations within industries classified by our methodology should be less than those within industries classified by other criteria. Moreover, if our methodology correctly classifies firms belonging to different industries into their respective industries, the across variations calculated by our method should be greater than those of other methodologies. Therefore, we verify the effectiveness of our proposed method by comparing the standard deviations of profitability and market risk within or across industries.

For the across industry variation, we first compute the weighted mean values of firm characteristics within the same industry. Then, we compute the standard deviation of the weighted mean values across all industries. For the within-industry variation, we first compute the standard deviation of firm characteristics in each industry. Then, we compute the industry-size weighted averages of the standard deviations. We expect the proposed method to have higher across-industry variation and lower within-industry variation than the other classification methods such as the three-digit SIC codes, GICS sub-industries, four-digit NAICS codes, and TNIC code. We represent profitability by dividing operating income (OI) or net income (NI) by asset size or sales: OI/asset, OI/sale, NI/asset, and NI/sale. We also use the market beta calculated from daily stock returns during each fiscal year to represent firm risk.

Table 4 reports the comparison results. Panel A (B) represents the across-industry (within-industry) variations of the variables. The first three rows (1–3) show the results of comparing our methodology with the existing fixed industry classifications. The numbers in the second column represent the number of industries classified by each classification method. For example, the firms in our sample are classified into 248 industries by the three-digit SIC codes, 174 by the sub-industry of GICS, and 243 by the four-digit NAICS codes. The number of industries in rows 4 through 7 is 300, which is the number of industries based on the text-based industry classification in Hoberg and Phillips (2016). Moreover, rows 4 through 7 represent the results of the text-based classifications. In particular, rows 4 and 6 are the results of Hoberg and Phillips's (2016) methodology, and rows 5 and 7 are those of our autoencoder-based industry classifications. Moreover, the results in row 4 are based on the 300 fixed industry classifications calculated from all of the word-of-bags, and those in row 5

Industry classification systems	Number Industries	OI/ Sale	OI/ Asset	NI/ Sale	NI/ Asset	Market Beta
A. Across-Industry standard deviations: firm-size weighted results						
1. SIC 3-digit industries	248	0.358	0.067	0.355	0.060	0.300
2. GICS subindustries	174	0.171	0.064	0.158	0.066	0.281
3. NAICS 4-digit industries	243	0.516	0.091	0.518	0.092	0.320
4. TNIC 300 fixed industries	300	1.433	0.089	1.476	0.087	0.337
5. Autoencoder + SKmenas	300	3.299	0.107	3.351	0.101	0.307
6. Original Intransitive TNIC	300	3.212	0.121	3.232	0.121	0.276
7. Autoencoder + Intransitive TNIC	300	2.640	0.105	2.562	0.104	0.254
B. Within-Industry standard deviations: industry-size weighted results						
1. SIC 3-digit industries	248	2.937	0.137	2.886	0.140	0.400
2. GICS subindustries	174	2.546	0.138	2.509	0.143	0.379
3. NAICS 4-digit industries	243	3.346	0.141	3.201	0.146	0.406
4. TNIC 300 fixed industries	300	2.514	0.140	2.560	0.145	0.402
5. Autoencoder + SKmenas	300	1.227	0.104	1.216	0.105	0.354
6. Original Intransitive TNIC	300	3.206	0.133	3.110	0.137	0.422
7. Autoencoder + Intransitive TNIC	300	2.538	0.130	2.452	0.136	0.422

Table 4. Across- and within-industry variations by industry classification systems

represent the results from the classification method based on the 10 reduced features extracted from the autoencoder. Next, the results in rows 6 and 7 are similar to those in rows 4 and 5, but these consider the possibility that one firm may belong to multiple industries at the same time. That is, rows 6 and 7 randomly select one firm, find firms with similar cosine similarities, and repeat this selection process until 300 industries are formed. Meanwhile, rows 4 and 5 simply classify our firms to belong to one of the 300 industries. Therefore, the last four rows of each panel investigate whether the industry classification can be improved when an autoencoder is applied (rows 5 and 7) by comparing row 4 against row 5 or row 6 against row 7.

Panel A of Table 4 shows the results of the across-variation comparison. The results show that the text-based industry classification method can taxonomize distinctive industries better than traditional fixed industry classifications can. For example, the across-industry variation for OI/Sale is 0.358, 0.171, and 0.516 for the three-digit SIC, GICS sub-industries, and four-digit NAICS, respectively, while that of the TNIC 300 fixed code is 1.433. These results imply that the text-based method clearly improves the classification in terms of cross-industry variations. When we apply the dimensionality reduction technique using an autoencoder to compute the spherical *k*-means, the variation is 3.299 (row 5), which is 1.3 times larger than that of the original TNIC 300 fixed code. We can find similar patterns in the results of NI/sale and other measures of profitability, even if the magnitudes of the variations are small in the OI/asset or NI/asset values. Moreover, we observe similar improvements in the market beta.

We confirm that considering the business description to determine each firm's industry provides better information than conventional fixed industry classifications. We also find that applying the autoencoder can improve the performance of TNIC based on across-industry variations of profitability in rows 4 and 5. The results in rows 6 and 7 show that applying the autoencoder may not improve the classification method. However, this result does not necessarily imply that autoencoders cannot improve the classification because the industry formation based on repeated selections should indicate that one firm might be selected into multiple industries. Therefore, reducing the dimension of word vectors by applying the autoencoder can improve the text-based industry classification approach in terms of cross-industry variations, at least when we categorize one firm into one industry.

Panel B of Table 4 shows the results of the within-industry variation. Similar to the results of the cross-industry variation, the results of the within-industry variations also show that text-based industry classification systems are an effective way to group homogeneous firms in terms of firm characteristics. We find that the within-industry variations for text-based classifications (rows 4–7) are much lower than those for conventional fixed industry codes (rows 1–3). For example, the within-industry variation of OI/sales is 2.937, 2.546, and 3.346 for three-digit SIC codes, GICS sub-industries, and four-digit NAICS, respectively, while that for the TNIC 300 fixed code is 2.514. We also find similar patterns throughout the columns when we calculate the within-industry variations by using other measures of profitability.

Moreover, we confirm that reducing the dimension of word vectors by applying an autoencoder improves the informativeness of the text-based industry classification approach. When we use the autoencoder and spherical *k* means, the within-industry variation is reduced to 1.227, which is 51.19% smaller than the within-industry variation of the original TNIC 300 code. These results are maintained even though we construct industries by choosing firms repeatedly in rows 6 and 7. Furthermore, we find that the within-industry variations for the market beta slightly decrease when we use dimensionally reduced word vectors by applying the autoencoder. That is, we find that the within-industry variations calculated using different measures of profitability, as well as the market beta, are lower in text-based classifications and much lower when using the autoencoder than when conventional fixed industry classifications are applied. Therefore, we conclude that overall, reducing the dimension of the vector space increases the informativeness of the text-based industry classification system.

4.4 Curse of high dimensionality

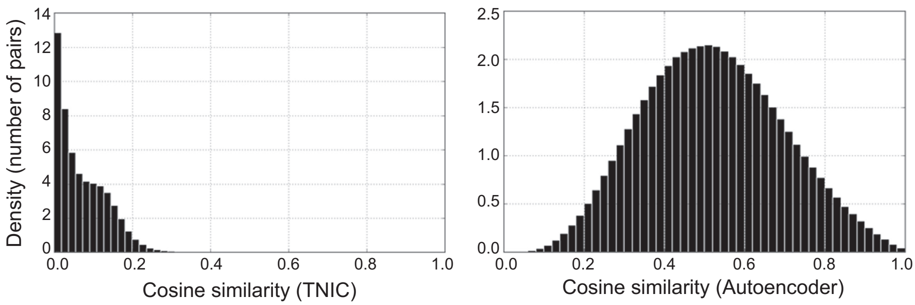
Radovanović *et al.* (2010) show that as dimensionality increases, the ratio between the standard deviation and the mean of the pairwise cosine similarities may converge to zero. The result implies that the pairwise cosine similarity measures approach a constant value, and its variance shrinks as dimensionality increases. Moreover, Aggarwal *et al.* (2001) show that the distance measure may not be qualitatively meaningful in high-dimensional space.

Table 5 shows the summary statistics of cosine similarities by applying different methods. We compute the cosine similarities by either using the word vectors from whole unique words or using the reduced word vectors extracted from the autoencoder. We use the cosine similarities of the original TNIC from the database provided by Hoberg and Phillips (2016). The mean and standard deviation of the cosine similarity for the original TNIC approach are 0.073 and 0.063, respectively, while those for using the autoencoder are 0.521 and 0.173, respectively. By computing the ratio between the mean and the standard deviation, the ratios for the original TNIC and for using the autoencoder are 1.16 and 3.01, respectively. Based on the argument of Radovanović *et al.* (2010), these statistics reveal that the ratio of the original TNIC is small, and thus may fail to distinguish one firm from another because the dimensions of the word vectors are very large.

Figure 6 illustrates the distribution of cosine similarities by applying different methods. The left panel reflects the distribution when using the original TNIC, and the right panel shows that when using the autoencoders. The left panel shows that the cosine similarities of all firm-pairs using whole words are skewed under the value of 0.2. The illustration from the left panel indicates that cosine similarities based on high dimensions may not distinguish firms well because the cosine similarities may have very similar values. However, the right panel shows that the cosine similarities are symmetrically and evenly distributed around the mean. This illustration suggests that our dimension reduction technique can enhance the variation among measures that could successfully distinguish one firm from another.

Table 5. Cosine similarities of the original TNIC and the applied autoencoder of the TNIC

Industry classification systems	Mean	SD	Min	Max
Original intransitive TNIC	0.073	0.063	0.000	0.904
Autoencoder + Intransitive TNIC	0.521	0.173	0.011	0.988



Note(s): The left figure represents the distribution of cosine similarities based on the TNIC database. The right figure represents the distribution of cosine similarities based on the dimension reduction output of the autoencoder

Figure 6. Comparison of cosine similarity measures

Therefore, we conclude that applying the autoencoder to high-dimensional word vectors can effectively mitigate the curse of high dimensionality.

5. Conclusions

In this paper, we present an alternative method to enhance existing industry classifications. While Hoberg and Phillips (2016) propose a text-based network industry classification based on 10-K business descriptions, we use a text mining technique to resolve the dimensionality problems in high-dimensional texts. We employ a machine learning technique to avoid the curse of dimensionality. In particular, we use the autoencoder to extract ten or two features that can explain the original input vector without loss.

A visual representation of the clustered firms shows that firms in similar industries are well-clustered. This result indicates that firms grouped in the same industry can have a similar radius with different vector norms as the feature of the cosine similarity measure. The classification results also show that the similarity and closeness between industries are well represented.

We also show that text-based classification is better than the existing industry classification systems. Moreover, our methodology can improve the existing text-based industry classification by relaxing the curse of dimensionality. Finally, we verify the effectiveness of our method by comparing the standard deviations of profitability and market risk within or across industries. We confirm that considering texts in the business descriptions to determine each firm's industry is more informative than traditional industry classifications. Moreover, reducing the dimension by applying the autoencoder can improve the previous text-based classification.

Our work contributes to the industry classification literature by empirically investigating the effectiveness of the text mining method. We find that product or service descriptions in firms' annual reports are useful for industry classification. The text mining method resolves issues concerning the timeliness of traditional industry classifications by capturing new information in annual reports. In addition, our approach using autoencoders can solve the computing concerns of high dimensionality. The proposed method provides both theoretical and practical implications for industry classification.

Nevertheless, our study has some limitations, in consideration of which future work may be conducted. First, we applied our method to a specific setting. We may secure generalizability by testing the machine learning method in various geographical and chronological contexts. We expect that learning processes will gradually improve the text mining method. Second, we input the business descriptions in the 10-K reports. If we add additional information, for example, the risk factors contained in other parts of the reports and footnotes, we may find more opportunities to advance the method. To enhance the proposed method, we suggest future studies to consider analyzing unstructured data in financial reports.

Note

1. For more information see www.fasb.org/pds/fas131.pdf.

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